Version 7.4.8

# PVsyst - Simulation report

## Grid-Connected System

Project: Qom-1MW

Variant: New simulation variant

Sheds on ground

System power: 968 kWp

Qom - Iran

**Author**

**PVsyst V7.4.8**
VC0, Simulation date:
10/01/26 19:37
with V7.4.8

## Project summary

**Geographical Site**
**Qom**
Iran

**Situation**

| | |
|---|---|
| Latitude | 34.65 °N |
| Longitude | 50.90 °E |
| Altitude | 928 m |
| Time zone | UTC+3.5 |

**Project settings**

| | |
|---|---|
| Albedo | 0.20 |

**Weather data**
Vādī os Salām
Meteonorm 8.1 (1988-2000), Sat=100% - Synthetic

## System summary

**Grid-Connected System**

**Sheds on ground**

**PV Field Orientation**
Fixed plane
Tilt/Azimuth 34 / 10 °

**Near Shadings**
Linear shadings : Fast (table)

**User's needs**
Unlimited load (grid)

**System information**

**PV Array**

| | |
|---|---|
| Nb. of modules | 1344 units |
| Pnom total | 968 kWp |

**Inverters**

| | |
|---|---|
| Nb. of units | 7 units |
| Pnom total | 875 kWac |
| Pnom ratio | 1.106 |

## Results summary

| | | | | | |
|---|---|---|---|---|---|
| Produced Energy | 1889231 kWh/year | Specific production | 1952 kWh/kWp/year | Perf. Ratio PR | 84.28 % |

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## General parameters

**Grid-Connected System**

**Sheds on ground**

**PV Field Orientation**

| Orientation | |
|---|---|
| Fixed plane | |
| Tilt/Azimuth | 34 / 10 ° |

| Sheds configuration | | |
|---|---|---|
| Nb. of sheds | 56 | units |
| **Sizes** | | |
| Sheds spacing | 10.5 | m |
| Collector width | 4.79 | m |
| Ground Cov. Ratio (GCR) | 45.6 | % |
| Top inactive band | 0.02 | m |
| Bottom inactive band | 0.02 | m |
| **Shading limit angle** | | |
| Limit profile angle | 22.4 | ° |

| Models used | |
|---|---|
| Transposition | Perez |
| Diffuse | Perez, Meteonorm |
| Circumsolar | separate |

**Horizon**
Free Horizon

**Near Shadings**
Linear shadings : Fast (table)

**User's needs**
Unlimited load (grid)

## PV Array Characteristics

**PV module**

| | |
|---|---|
| Manufacturer | Mana Energy |
| Model | MEP720-132-GB |
| (Custom parameters definition) | |
| Unit Nom. Power | 720 Wp |
| Number of PV modules | 1344 units |
| Nominal (STC) | 968 kWp |
| Modules | 56 string x 24 In series |
| **At operating cond. (50°C)** | |
| Pmpp | 899 kWp |
| U mpp | 943 V |
| I mpp | 953 A |

**Inverter**

| | |
|---|---|
| Manufacturer | Sungrow |
| Model | SG125-HV |
| (Original PVsyst database) | |
| Unit Nom. Power | 125 kWac |
| Number of inverters | 7 units |
| Total power | 875 kWac |
| Operating voltage | 860-1450 V |
| Pnom ratio (DC:AC) | 1.11 |

**Total PV power**

| | |
|---|---|
| Nominal (STC) | 968 kWp |
| Total | 1344 modules |
| Module area | 4172 m² |
| Cell area | 1956 m² |

**Total inverter power**

| | |
|---|---|
| Total power | 875 kWac |
| Number of inverters | 7 units |
| Pnom ratio | 1.11 |

## Array losses

**Thermal Loss factor**

| Module temperature according to irradiance | |
|---|---|
| Uc (const) | 29.0 W/m²K |
| Uv (wind) | 0.0 W/m²K/m/s |

**DC wiring losses**

| | |
|---|---|
| Global array res. | 16 mΩ |
| Loss Fraction | 1.5 % at STC |

**LID - Light Induced Degradation**

| | |
|---|---|
| Loss Fraction | 1.3 % |

**Module Quality Loss**

| | |
|---|---|
| Loss Fraction | -0.2 % |

**Module mismatch losses**

| | |
|---|---|
| Loss Fraction | 2.0 % at MPP |

**Strings Mismatch loss**

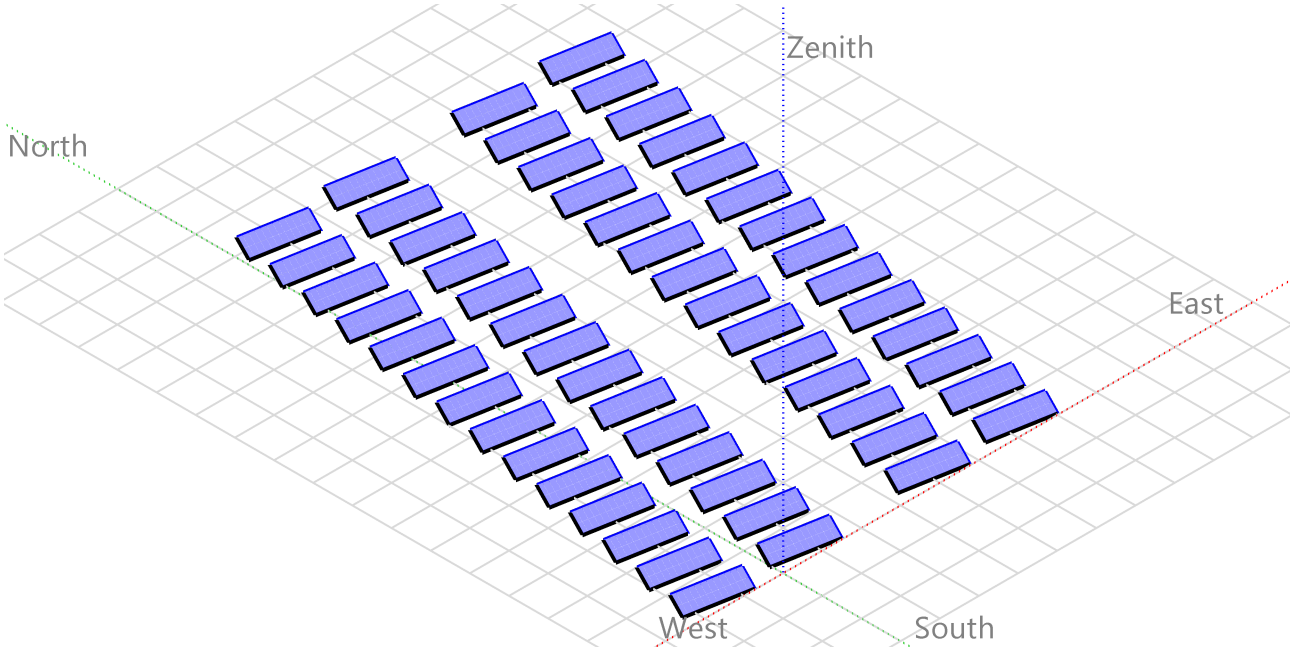
| | |
|---|---|
| Loss Fraction | 0.1 % |

**IAM loss factor**
Incidence effect (IAM): Fresnel smooth glass, n = 1.526

| 0° | 30° | 50° | 60° | 70° | 75° | 80° | 85° | 90° |
|---|---|---|---|---|---|---|---|---|
| 1.000 | 0.998 | 0.981 | 0.948 | 0.862 | 0.776 | 0.636 | 0.403 | 0.000 |

## Near shadings parameter

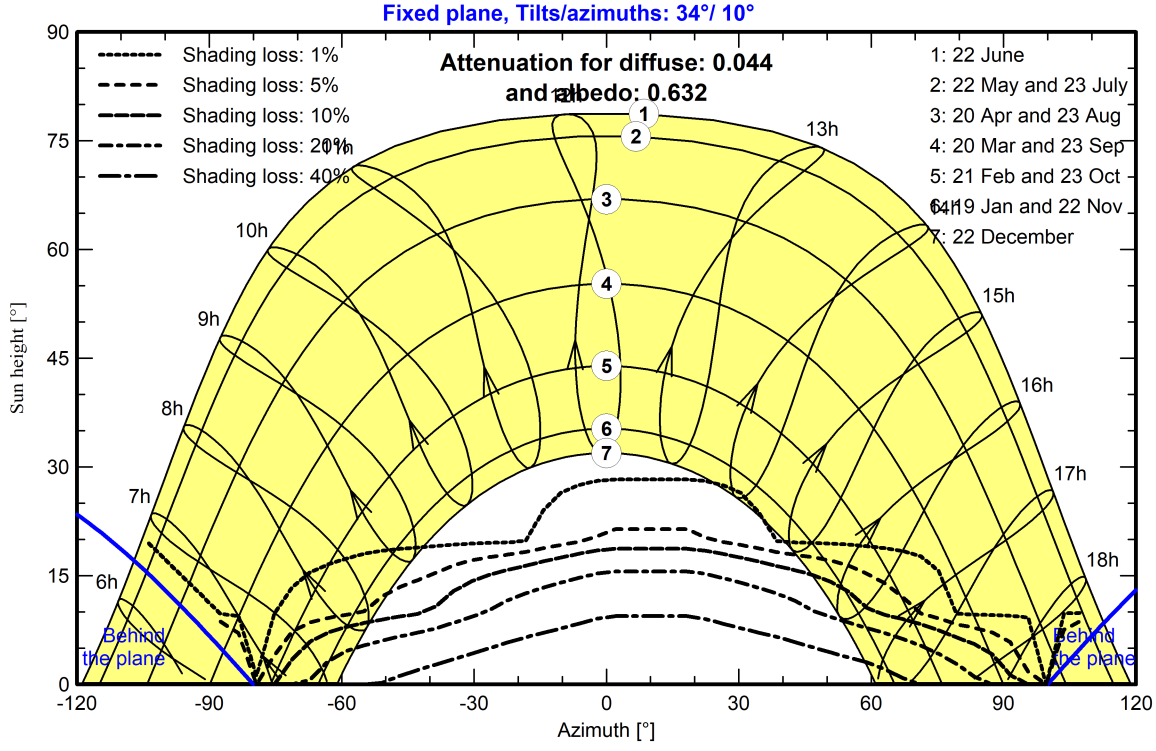
**Perspective of the PV-field and surrounding shading scene**




## Iso-shadings diagram

**Orientation #1**

**Fixed plane, Tilts/azimuths: 34°/ 10°**



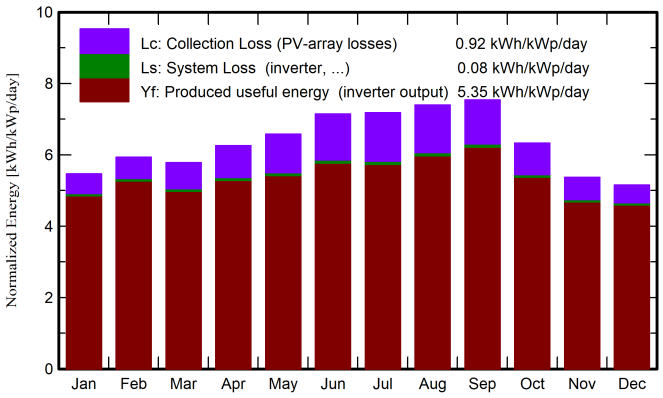
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## Main results

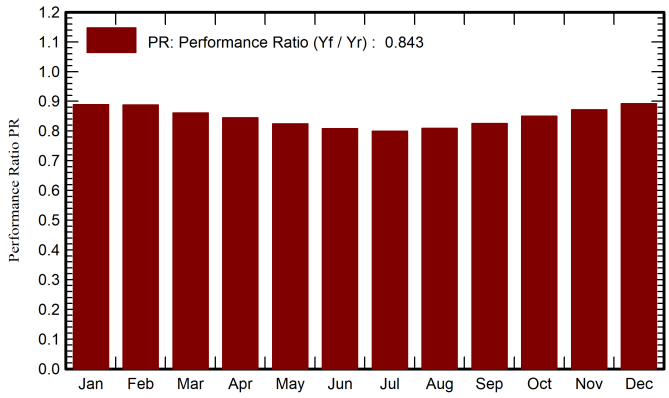
### System Production

| | | | |
|---|---|---|---|
| Produced Energy | 1889231 kWh/year | Specific production | 1952 kWh/kWp/year |
| | | Perf. Ratio PR | 84.28 % |

**Normalized productions (per installed kWp)**




**Performance Ratio PR**




### Balances and main results

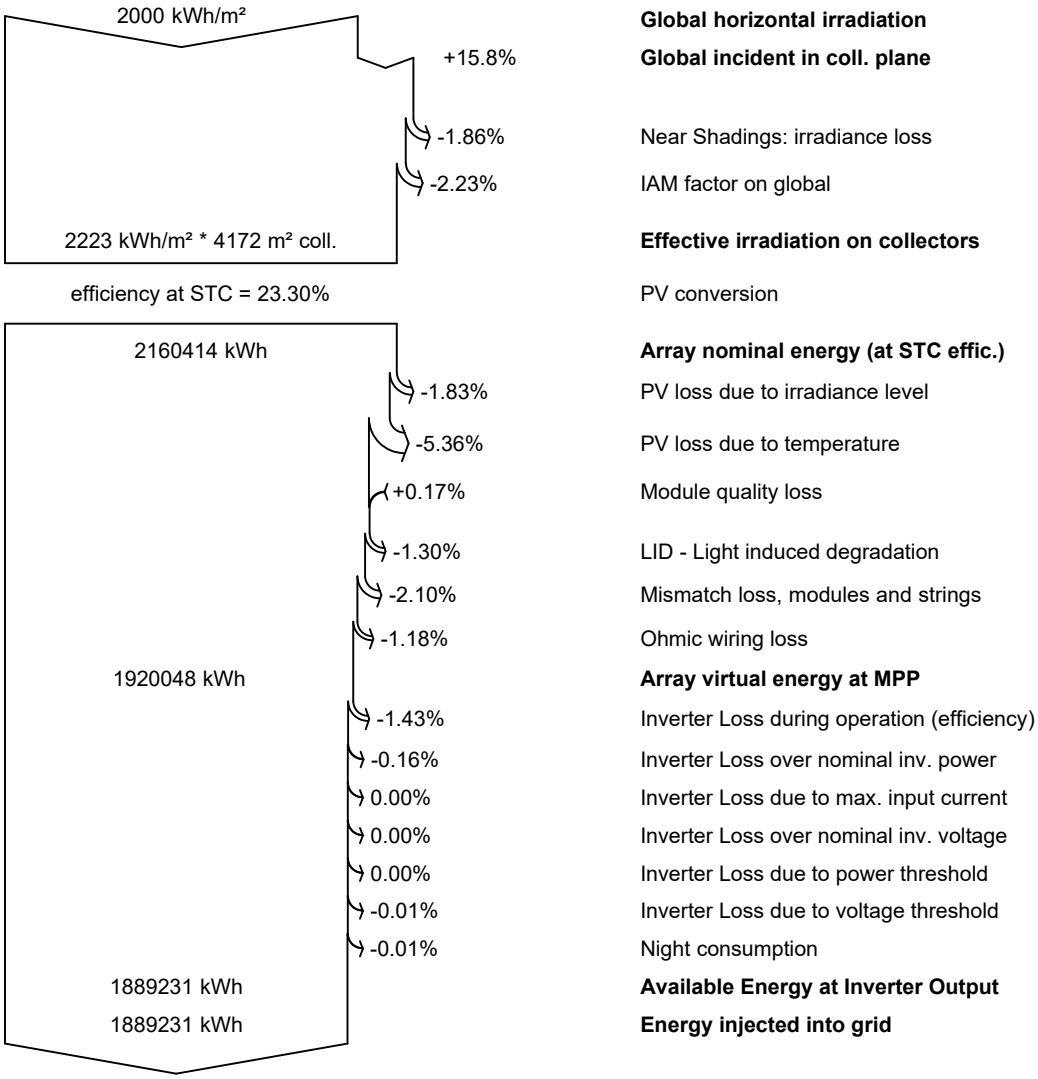
| | GlobHor kWh/m² | DiffHor kWh/m² | T_Amb °C | GlobInc kWh/m² | GlobEff kWh/m² | EArray kWh | E_Grid kWh | PR ratio |
|---|---|---|---|---|---|---|---|---|
| **January** | 102.0 | 26.77 | 5.11 | 169.4 | 162.4 | 147720 | 145735 | 0.889 |
| **February** | 116.1 | 33.22 | 7.98 | 166.1 | 160.7 | 144698 | 142719 | 0.888 |
| **March** | 149.8 | 57.49 | 13.89 | 179.4 | 172.1 | 151708 | 149462 | 0.861 |
| **April** | 178.7 | 66.78 | 18.94 | 187.8 | 179.6 | 155811 | 153489 | 0.844 |
| **May** | 214.2 | 73.20 | 25.50 | 204.0 | 194.4 | 165222 | 162618 | 0.824 |
| **June** | 238.8 | 68.10 | 30.84 | 214.5 | 204.5 | 170272 | 167678 | 0.808 |
| **July** | 241.3 | 67.65 | 34.12 | 222.7 | 212.6 | 174823 | 172163 | 0.799 |
| **August** | 226.1 | 54.26 | 32.31 | 229.3 | 220.1 | 181996 | 179423 | 0.809 |
| **September** | 193.6 | 35.38 | 27.01 | 226.1 | 218.0 | 183101 | 180637 | 0.826 |
| **October** | 144.9 | 37.72 | 20.32 | 196.2 | 189.7 | 163576 | 161348 | 0.850 |
| **November** | 102.2 | 29.63 | 11.28 | 161.2 | 154.7 | 137946 | 135991 | 0.872 |
| **December** | 92.5 | 23.77 | 6.47 | 159.8 | 153.8 | 139933 | 137966 | 0.892 |
| **Year** | 2000.4 | 573.97 | 19.55 | 2316.4 | 2222.6 | 1916805 | 1889231 | 0.843 |

### Legends

| | |
|---|---|
| GlobHor | Global horizontal irradiation |
| DiffHor | Horizontal diffuse irradiation |
| T_Amb | Ambient Temperature |
| GlobInc | Global incident in coll. plane |
| GlobEff | Effective Global, corr. for IAM and shadings |
| EArray | Effective energy at the output of the array |
| E_Grid | Energy injected into grid |
| PR | Performance Ratio |

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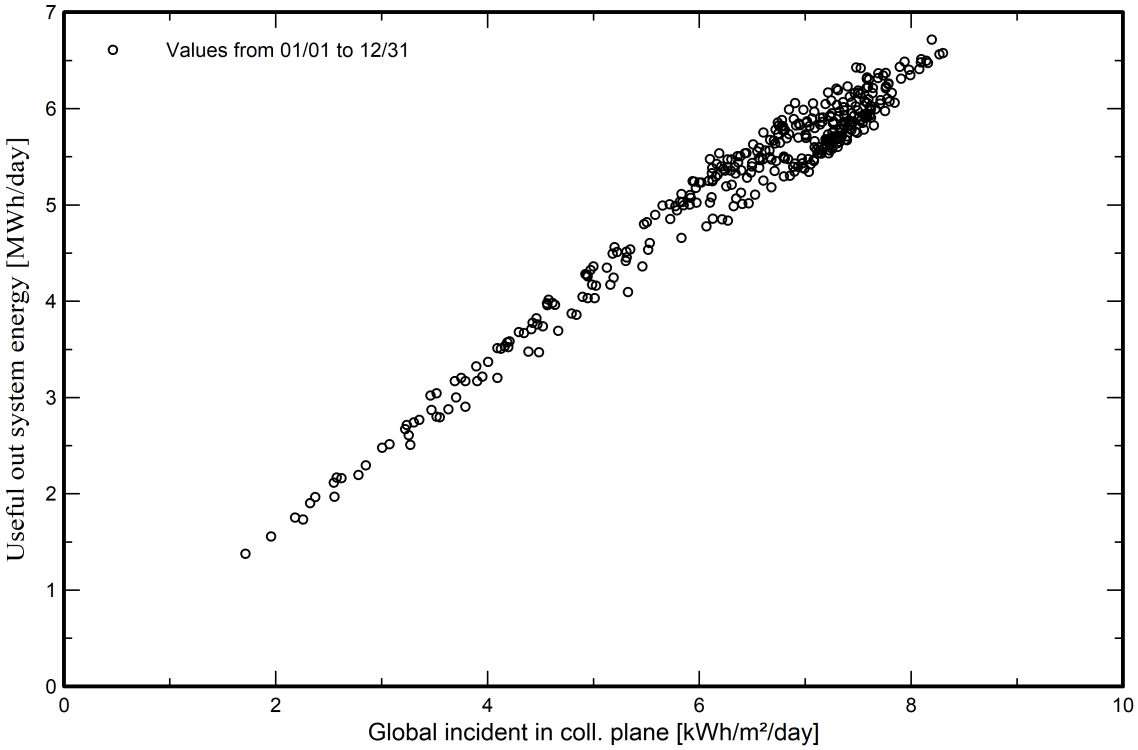
## Loss diagram



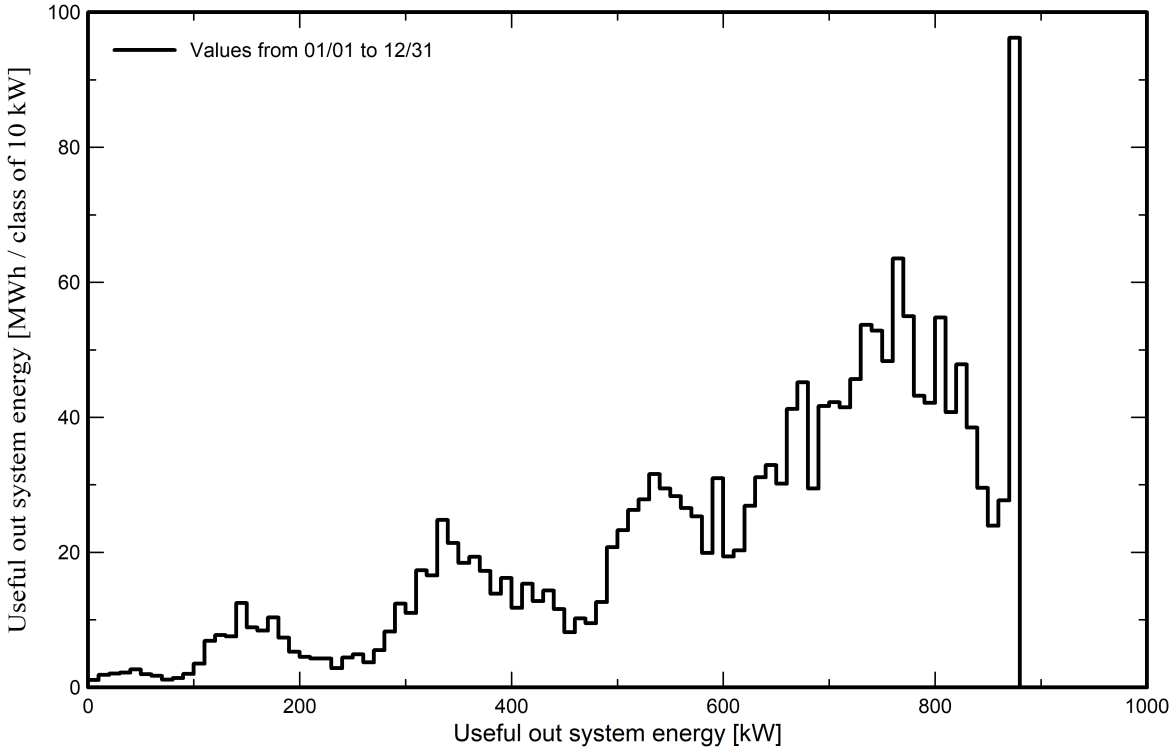
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## Predef. graphs

### Daily Input/Output diagram

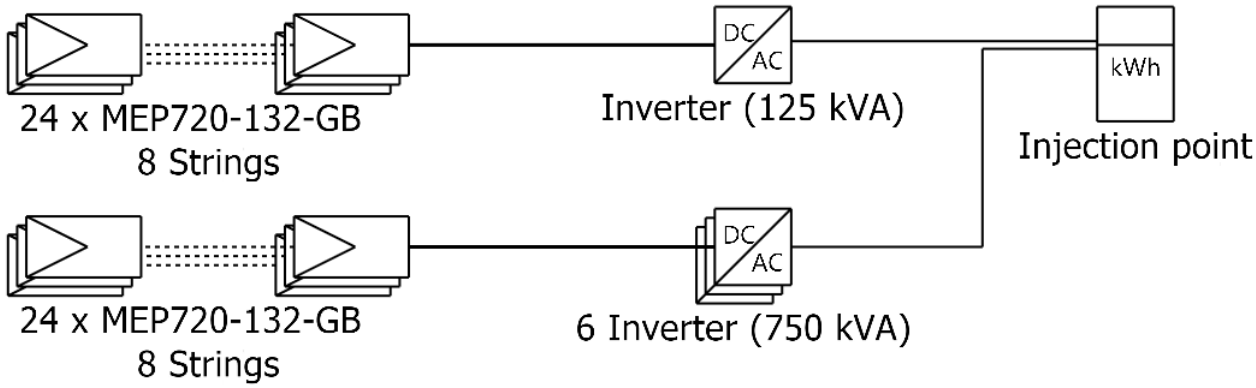



### System Output Power Distribution



# Single-line diagram

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| | |
|---|---|
| PV module | MEP720-132-GB |
| Inverter | SG125-HV |
| String | 24 x MEP720-132-GB |

| | | |
|---|---|---|
| | Qom-1MW | |
| | VC0 : New simulation variant | 10/01/26 |